

Mathematics Reference: Mixture and Alligation

Educational Series

December 29, 2025

1 Introduction

- **Mixture:** The process of mixing two or more ingredients of different values together to produce a product of a specific intermediate value.
- **Alligation:** A rule that enables us to find the ratio in which two or more ingredients at their respective prices must be mixed to produce a mixture of a desired price.

2 The Alligation Rule

If two ingredients are mixed, then:

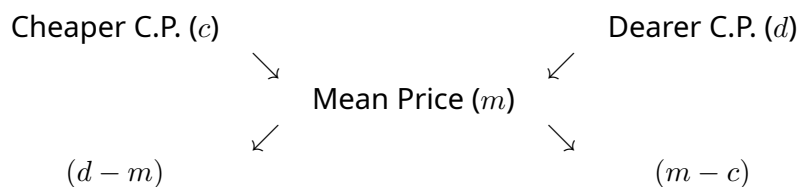
$$\frac{\text{Quantity of Cheaper Ingredient}}{\text{Quantity of Dearer Ingredient}} = \frac{\text{Cost Price of Dearer} - \text{Mean Price}}{\text{Mean Price} - \text{Cost Price of Cheaper}}$$

2.1 Visual Representation (Alligation Cross)

Let:

- c = Cost price of cheaper ingredient
- d = Cost price of dearer ingredient
- m = Mean price (price of the mixture)

The ratio of Cheaper to Dearer is $(d - m) : (m - c)$.



3 Mean Price Formula

The Mean Price (m) is the cost price of a unit quantity of the mixture. If q_1 quantity of value c is mixed with q_2 quantity of value d :

$$m = \frac{q_1 c + q_2 d}{q_1 + q_2}$$

4 Repeated Dilutions (Replacement Formula)

Suppose a container contains x units of a liquid (e.g., pure milk) from which y units are taken out and replaced by water. After n such operations:

$$\text{Quantity of pure liquid remaining} = x \left(1 - \frac{y}{x}\right)^n$$

5 Important Notes

- The Alligation Rule can only be applied when all values (c, d, m) are of the **same unit** (e.g., all must be Cost Prices).
- If the selling price of the mixture is given, you must first calculate the Mean Cost Price using Profit/Loss formulas before applying alligation.

6 Solved Example

Problem: In what ratio must rice at \$62 per kg be mixed with rice at \$72 per kg so that the mixture is worth \$65 per kg?

Solution:

- Cheaper C.P. (c) = 62
- Dearer C.P. (d) = 72
- Mean Price (m) = 65
- Ratio = $(d - m) : (m - c)$
- Ratio = $(72 - 65) : (65 - 62) = 7 : 3$

The two types of rice must be mixed in the ratio **7:3**.